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Electrothermal Modeling of Lithium-Ion Batteries for Electric Vehicles

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***Abstract*—The performance of lithium-ion batteries is contingent on the operational temperature and applied current rate. Especially in electric vehicle (EV) application, repeated heavy-duty drive cycles impose intense thermal stress on batteries, resulting in safety concerns. This article explains the electrochemistry within the battery and analyzes the dependence of battery internal parameters on temperature and current rate based on their electrochemical nature. In addition, this interdependence is used in the equivalent circuit model to predict the battery behavior under different drive cycles. Finally, the behavior under different drive cycles is analyzed to improve the battery performance in electric vehicles.**

Index Terms—Lithium-ion battery; electrochemistry; thermal; drive cycles.

TABLE I
Nomenclature

Parameter	Description
A_s	Surface area of the cell, m^2
A_f	Frontal area of the vehicle, m^2
C_d	Air friction coefficient
C_{dl}	Double layer capacitance, F
C_P	Density of the cell, kg/m^3
c_2	Liquid-phase salt concentration, mol/m^3
E_{act}	Activation energy for the reaction, J/mol
g	Gravitational acceleration, $9.8 m/s^2$
i	Applied current, A
i_{dl}	Current through the double layer capacitance, A
i_e	Liquid-phase current density, A/m^2
J	Local electrode current density, A/m^2
M	Mass of the vehicle, kg
m	Mass of the cell, kg
P	Power, W
Q	Capacity, Ah
$\dot{Q}_{gen}$	Rate of heat generation, W
$\dot{Q}_R$	Rate of resistive heat generation, W
$\dot{Q}_E$	Rate of entropy heat generation, W
$\dot{Q}_C$	Rate of heat convection, W
R_O	Ohmic resistance, Ω
R_{ct}	Charge transfer resistance, $\Omega \cdot m^2$
$\bar{R}$	Universal gas constant, 8.31 J/(mol·K)
T	Temperature, K
T_A	Ambient temperature, K
T_{ref}	Reference temperature, K
t	Time, s
U_{OC}	Open circuit voltage (OCV), V
V	Velocity of the vehicle, m/s
V_D	Voltage across the RC circuit, V
V_t	Terminal voltage, V
Greek	
α	Slope of a road, rad
β	Temperature coefficient of resistivity
ε	Volume fraction of a phase
η	Local overpotential driving electrochemical kinetics, V
η_M	Motor Efficiency
κ	Ionic conductivity of electrolyte, S/m
κ_D	Diffusional conductivity of electrolyte, Am^2/mol
μ	Rolling resistance
ρ	Air density, kg/m^3
σ	Electrical conductivity, S/m
ρ_1	Electrical resistivity, Ωm
τ	Electrochemical time constant
Φ	Temperature dependent parameter

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I. Introduction

Lithium-ion batteries (LIB) present a promising solution for powering electric vehicles since they feature high energy density, power density, low self-discharge and more importantly, long cycle life. However, under heavy-duty operation in EV applications, thermal behavior becomes a more delicate issue in battery management. In general, thermal behavior of a cell is influenced by two factors: the internal factor, namely battery internal parameters and the external factor, elicitation of drive cycles. These two factors determine heat generation within a battery cell, causing temperature rise and in return the temperature variation may influence battery internal parameters. In long term, the coupled electrothermal relationship defines the operating environment of the battery cell, and furthermore, the cycle life.

The battery internal parameters are approximated based on the electrochemistry within the cell. Researchers have spent years in understanding the electrochemistry so that the parameters can be modeled in an accurate manner [1-8]. In

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particular, the models based on Porous Electrode Theory proposed by Doyle et al. [1] and Single Particle Model by Haran et al. [2] have been widely used for modeling the dynamic response of LIBs. All these models show that temperature and current rate are highly involved in the battery dynamics, such as interfacial charge transfer kinetics [4], [5] and electrolyte kinetics [6], [8], [9].

While electrochemical models have shown the ability to accurately predict the concentration dynamics and terminal voltage, they are characterized by coupled partial differential equations, which overburdens the computation when applied to fast simulation or control design. For this reason, equivalent circuit models are used to simulate the electrical behavior. In 2004, Plett proposed and compared a series of lumped models [10]. Consequently, equivalent circuit models with resistance-capacitance (RC) circuit were used to describe the battery dynamics, including the first-order RC model [11-13], second-order RC model [14], [15] and third-order RC models [16]. Moreover, different derivatives are proposed to simulate more details in the dynamics, such as hysteresis effects [17], Warburg impedance [18] and aging effects [19]. In this paper, the focus is on the dependence of the internal parameters on temperature and current rate in the context of their electrochemical nature. Specifically, this paper will explain how temperature and current rate influence the electrochemical process within the lithium-ion cell and furthermore their impact on the variation of battery internal parameters.

In addition, different models and methods have been proposed to analyze or predict the thermal behavior of LIBs [20-27]. These models could be electrochemical models [20-22], [25], empirical models [24] or equivalent circuit models [23], [26]. Subsequently, the thermal behavior of the cell is analyzed, using Finite Element method [25], [27] or other numerical methods [20-24]. The previous work is rather profound in the energy conversion inside the battery, yet it did not consider the dependence of parameters on temperature and current rate. Thus, they are not suitable for EV application with heavy-duty drive cycles.

In the case of drive cycles, the cell starts to generate heat due to the electrochemical reactions. Subsequently, the temperature variation caused by heat generation will act on the electrochemical behavior and further influence the thermal behavior. Although, there are several papers which tried to discuss this process [17], [18], [24], [25], [26], [28], [29], they did not consider the detailed electrochemistry behind the effects of temperature and current rate. With detailed electrochemistry explained, this paper includes the effects of temperature and current rate in the modeling and analysis of lithium-ion cells therefore giving a more accurate prediction on the electro-thermal behavior of the cell. In this paper, the major contributions can be concluded as follows: 1. The effects of temperature and current rate on the battery internal parameters have been presented and explained based on the electrochemical principles within the lithium-ion battery. 2. The model proposed for the study includes these effects and the sensitive coupling relationship between the electrochemical and thermal behaviors of the cell. 3. Finally, the model is applied to the battery cell and the resultant thermal behavior under different drive cycles is compared for the purpose of improvement in battery performance.

This paper is organized as follows. Section II introduces the battery model and the experimental procedures for cell characterization. Section III discusses in detail the dependence of model parameters on temperature and current rate. Section IV explores the dependence relationship as applied in the model to validate the cell performance under different drive cycles. Finally, conclusions are presented in Section V.

II. Battery Modeling and Characterization

A. Battery Model

Fig. 1 shows the schematic of ion flowing between electrodes in an electrochemical battery cell. The negative or positive electrode coated with the porous solid-phase material is soaked in the electrolyte liquid/gel phase with the galvanic isolation of the separator in between. During discharge, lithium ions de-intercalate from the negative electrode, flowing through the highly conductive electrolyte, passing the separator and then intercalate into the positive electrode.

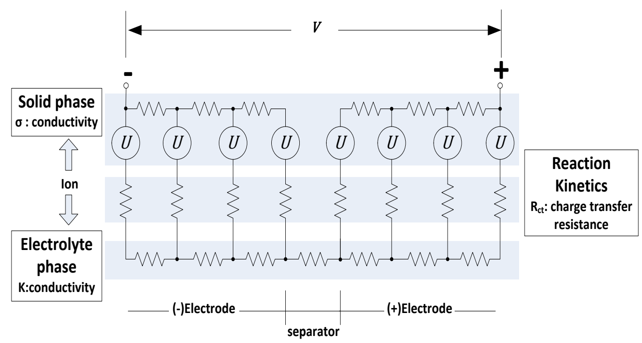

Fig. 1 Schematic of ion flow in a battery cell

At the electrodes, there is an open-circuit potential U that drives the ion flow. The solid and electrolyte phases have conductivity of σ and κ, respectively. For the solid phase, the conductivity σ follows Ohm's Law and is temperature dependent. Indeed, its reciprocal, resistivity ρ_1 follows:

$$\rho_1(\mathrm{T}) = \rho_{ref}[1 + \beta(T - T_{ref})] \tag{1}$$

For the electrolyte phase, the relationship between the solution potential and the current is given by [8]:

$$-\nabla\phi_e = \frac{i_e}{\kappa^{eff}} + \frac{\kappa_D^{eff}}{\kappa^{eff}}\nabla \ln c_2 \tag{2}$$

where $\kappa^{eff} = \kappa\varepsilon^{1.5}$ using the Bruggeman relation, $\kappa_D^{eff} = \kappa_D\varepsilon^{1.5}$. The temperature dependence of electrolyte ionic conductivity κ and electrolyte diffusional conductivity κ_D can be described by the Arrhenius equation:

$$\Phi = \Phi_{ref}\exp[\frac{E_{act}}{\bar{R}}(\frac{1}{T_{ref}} - \frac{1}{T})] \tag{3}$$

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Since current rate influences the concentration in the electrolyte, the concentration dependent coefficients κ, κ_D will also change accordingly [9].

Meanwhile, the electrode/electrolyte interfacial kinetics is described by the charge transfer process and will be approximated by the charge transfer resistance in this study. For small current density ($|J|, A/m^2$), the electrode reactions generate overpotential as follows:

$$\eta = R_{ct} J \tag{4}$$

For large current density ($|J|$), the overpotential is proportional to the natural logarithm of the current, where the proportionality constant is referred to as the Tafel slope [30]. The conductivity of the charge transfer ($1/R_{ct}$) also follows the Arrhenius Law [5]. Conclusively, the charge transfer resistance is contingent on current rate and temperature.

In parallel with the charge transfer resistance, a double layer capacitor often forms at the solid electrolyte interface (SEI) and produces a current [30] given by:

$$i_{dl} = C_{dl} \frac{\partial \eta}{\partial \mathrm{t}} \tag{5}$$

where C_{dl}(F/cm^2) is the double layer capacitance.

For the purpose of circuit simplification, the distributed structure in the above electrochemical analysis needs to be approximated into a concentrated model. Considering the connection between the electrodes, electrolyte and their interfaces, the equivalent circuit model in this study will herein be a first order model as shown in Fig. 2. In addition, it has been reported that there is Warburg impedance and high-frequency inductance in the equivalent circuit based on the observation from electrochemical impedance spectroscopy [18], [28]. The Warburg impedance is approximated to simulate the diffusion in solid and electrolyte phase of the cell, which dominates in the low-frequency range. In this study, the Warburg impedance is omitted for simplicity and the accuracy won't be tangibly comprised, as will be shown in Section IV. As for the inductance, it shows measurable impedance in the high-frequency range, which is out of the scope for this paper, since most EV applications operate in the range of DC to 10 Hz.

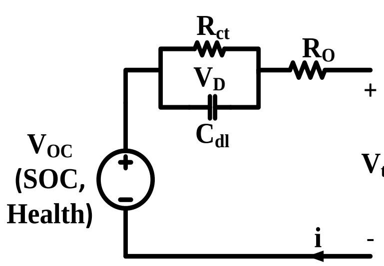

Fig. 2. Equivalent circuit of an electrochemical cell

In Fig. 2, the circuit consists of one ohmic resistance and one RC circuit. The RC circuit describes dynamic behavior in the battery due to the charge transfer resistance R_{ct} and the double layer capacitance C_{dl}. The dynamic equation that describes the voltage drop of the RC circuit is given by

$$\frac{dV_D}{dt} = -\frac{V_D}{C_{dl}R_{ct}} + \frac{1}{C_{dl}} i \tag{6}$$

where $\tau = C_{dl}R_{ct}$ is the electrochemical time constant of the system. Additionally, the state of charge (SOC) is given by using Coulomb counting as shown below

$$\mathrm{SOC} = \mathrm{SOC}_{init} - \frac{\int i dt}{Q} \tag{7}$$

Based on Kirchoff's Law, the terminal voltage can be expressed by

$$V_t = U_{OC} - iR_O - V_D \tag{8}$$

R_O represents the resistive effects of the electrodes, electrolyte and the contact resistance.

To determine the heat generation within the electrochemical battery, Bernadi explained the energy balance in 1985 [29]. Generally, the heat generation can be categorized into two parts, the resistive heat and the entropic heat and expressed by:

$$\dot{Q}_{gen} = \dot{Q}_R + \dot{Q}_E \tag{9}$$

$$= i(U_{OC} - V_t) + iT\frac{\partial U_{OC}}{\partial T} \tag{10}$$

where $\frac{\partial U_{OC}}{\partial T}$ is the entropy coefficient which is dependent on SOC. In this study, it will be calculated based on the dependence of OCV on temperature in Section III.

After calculating the heat generation within the battery, the temperature rise can be calculated by the equations of heat transfer and heat convection as follows

$$mC_P \frac{\partial T}{\partial t} = \dot{Q}_{gen} - \dot{Q}_C \tag{11}$$

$$\dot{Q}_C = hA_s(T - T_A) \tag{12}$$

B. Cell Characterization

To capture the electrical properties of batteries, two tests will be performed on batteries: static capacity test and hybrid pulse power characterization test;

The static capacity test measures the capacity of the battery using a constant discharge current rate. In this work, the battery is at the full charge state in the beginning and the discharge starts after an hour rest. The discharge is at a current rate of 0.3 C and lasts until the terminal voltage reaches the minimum limit of 3 V.

The hybrid pulse power characterization (HPPC) Test is used to determine the dynamic power capability over the applicable voltage range by applying both discharge and charge pulses. One typical pulse profile is shown in Fig. 3. Between each pair of discharge and charge pulses, the battery is discharged by 10% SOC increment to traverse the SOC range at a rate of 2 C. For each point of SOC, after the one-hour relaxation, the terminal voltage is eventually the open circuit voltage.

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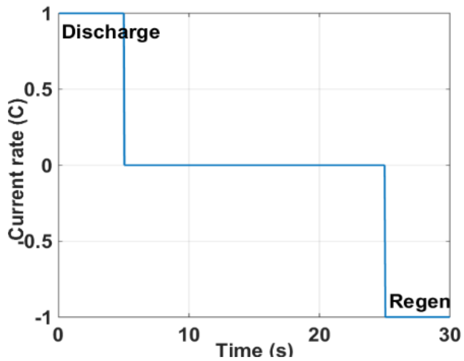


Fig. 3 A pair of discharge and charge pulse profile

III. Parameter Estimation using Experiment Results

A. Experimental Setup

The experimental setup for the study is shown in Fig. 4. It is comprised of a thermal chamber with the tested battery inside, four K-type thermocouples attached on the surface of the cell. A tester from Arbin Instruments is used to apply test profiles to the battery. The tester has four channels which feature independent control of current and voltage with accuracy at 0.05% full scale range (FSR), and then interfaces with the desktop for postprocessing.

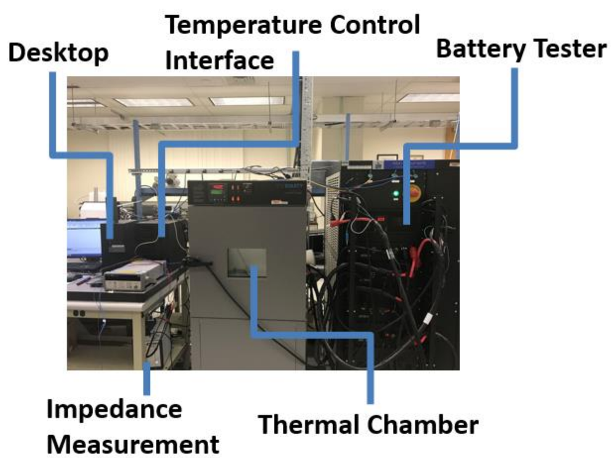


Fig. 4 Experimental setup

B. Data collection

The basic properties of the battery cell are shown in Table II. The model parameters in this study is dependent on operating conditions, including SOC, temperature and current rate. Based on (6)-(8), four parameters need to be identified: U_{OC}, R_O, R_D and C_D, to predict the electrical behavior of the cell. All these parameters are captured by running HPPC test profile multiple times at the corresponding temperatures and current rates. Detailed estimation method can be found in [32]. The SOC partition for the parameter is selected to be [0, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%]. The temperature partition is selected to be [0ºC, 10ºC, 20ºC, 30ºC, 40ºC]. Similarly, the partition for current rating is 0.8A, 1.6A, 3.2A, 4A and 7A. All data is collected at these operating points.

TABLE II
Basic properties of the battery cell

Parameter	Value	Parameter	Value
Nominal voltage (V)	3.75	Cathode material	$LiMn_2O_4$
Nominal capacity (Ah)	1.6	Cathode collector	Aluminum
Max. discharge current (A)	10	Anode material	Graphite
Mass (g)	43	Anode collector	Copper
Heat capacity (J/kg·K)	1013	Electrolyte	EC:DCM 1:1, $LiPF_6$

As reported in [8], LIBs will present abnormal performance at subzero conditions, such as capacity loss and voltage recovery and will show thermal hazard at temperature over 65ºC. The LIBs in these cases need to be handled with special care, and therefore the operating temperature ranges from 0 to 40ºC. Additionally, during the estimation process in HPPC test, the operating temperature, which is fixed by the thermal chamber, is considered the battery temperature, since after one-hour relaxation, the battery temperature reaches the steady state which is the ambient temperature. Because the tested battery is a thin pouched cell with a maximum temperature difference of 0.4 ºC on the surface at the steady state value, the battery will be viewed as a lumped model without considering the difference between the core and the surface temperature.

Meanwhile, current rate also affects the electrochemical properties of the cells. Considering the capacity of the cell being 1.6 Ah, the current partition [0.8 A, 1.6 A, 3.2 A, 4 A, 7 A] corresponds to [0.5 C, 1 C, 2 C, 2.5 C and 4.375 C], which covers from the low current rate to the high current rate. Thus, the pulse magnitude (charge and discharge) in HPPC test will be adjusted according to the current partition.

During the procedure, the actual SOC is estimated by Coulomb counting method, (the initial SOC must be calibrated). For each data collection, the battery starts from the state of full charge. Then the battery is soaked at the desired temperature for three hours followed by the test. Since the battery ability in storing charge is different at different temperatures, it is expected that the initial OCV of the battery to be different at different temperatures. However, it is still reasonable to keep the initial SOC as 100% so that the actual SOC during the test can be estimated.

C. Parameter Estimation

After these parameters (U_{OC}, R_O, R_D and C_D) are obtained, it is useful to reproduce these parameters for all the temperatures and current rates of interest.

Since the relationship between the inputs in this study, i.e. temperature, SOC and current rate, and the outputs, i.e. U_{OC}, R_O, R_D and C_D is nonlinear, the support vector method is selected. This method is generally applied to classify data groups by using hyperplanes. In this study, the estimation works similarly and the hyperplane here is used to categorize a subset of the input sample vectors which are deemed sufficient to derive the output in question.

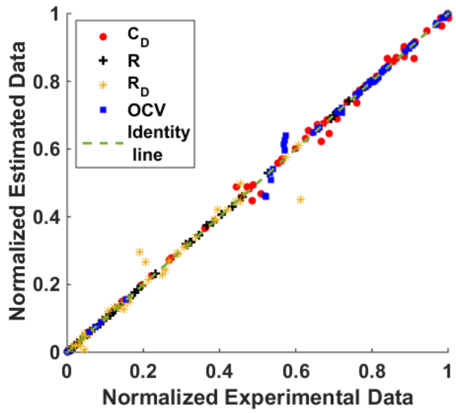


Fig. 5 Fitness between the experimental results and the predicted results

Fig. 5 shows the fitness between the experimental results and the predicted results. Both results are normalized for comparison. Notably, most of the predicted results show agreement with the corresponding experimental results.

D. Dependence of Parameters on Temperature

Per the Arrhenius Law, the chemical reaction rate is affected by temperature. It is reasonable to assume that the model parameters are contingent on temperature since the parameters are approximated by the electrochemical properties of the battery.

OCV is usually used to indicate the SOC of the battery and the value of OCV can represent the amount of charge that is left in the battery cell. Since the battery ability to store the charge is different at different temperatures, the OCV could be different even if the cells are at the same SOC. The profile of OCVs at different temperatures is shown in Fig. 6. The characteristic is used to calculate the entropy coefficient by (10). To focus on the effects of temperature and exclude the effects caused by current rates, the experiment in this section is conducted at a fixed current rate of 4 A.

R_O represents the resistive effects from the solid phase, the electrolyte phase and the other contact resistance attached on the cell and its dependence on temperature is shown in Fig. 7. From the figure 7 one can note that, the resistance of R_O decreases at all SOCs when the temperature increases. This conclusion was also found in [29]. Although the electrical resistivity of the solid phase and the contact resistance shows a positive correlation with the temperature based on (1), from the figure, the positive correlation is overridden by the inverse correlation between electrolyte conductivity (κ and κ_D) and temperature based on the Arrhenius Law. Based on (2), the electrolyte potential drop stems from two parts: ohmic loss and concentration overpotential. The concentration overpotential is caused by the diffusional ionic current, which can be ignored in diluted solutions, but may be dominant in high concentrated (>2.5 mol/L) solutions [9]. Since κ increases as temperature rises, it is expected that the resistive effect of the electrolyte decreases, leading to a decrease in R_O.

Fig. 8 shows the dependence of R_{ct} on temperature. It illustrates the interfacial kinetics between the electrode and the electrolyte. As mentioned before, the conductivity of the interfacial charge transfer ($1/R_{ct}$) also follows the Arrhenius Law. Consequently, the charge transfer resistance shows an inverse correlation with temperature. This result is in accordance with [29].

Fig. 9 shows the dependence of C_{dl} on temperature. The result can be supported by the results in [29]. At the solid electrolyte interface, charged particles of lithium ions aggregate and form a double layer, which displays the capacitive effect in parallel with the charge transfer resistance. As analyzed before, as the temperature increases, the ionization and diffusion effects increase, which means more lithium-ions accumulate at the interface. Thus, more charge gathers, which increases the capacitance.

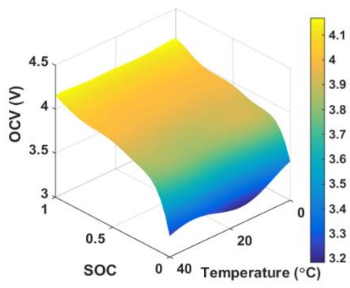
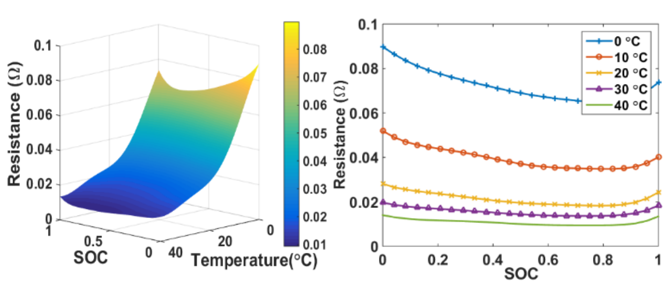


Fig. 6 Dependence of OCV on temperature



(a) (b)

Fig. 7 Dependence of R_O on temperature

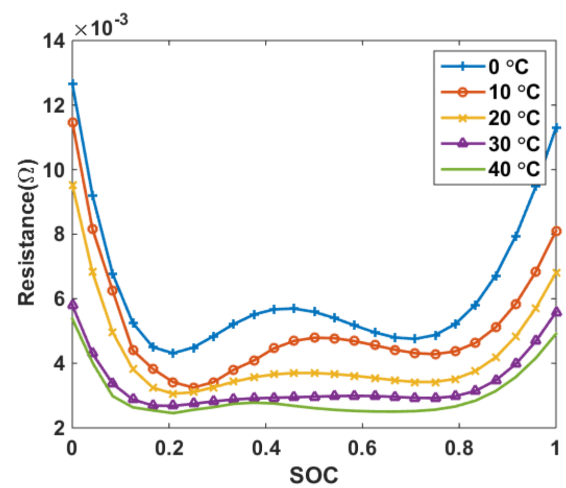


Fig. 8 Dependence of R_D on temperature

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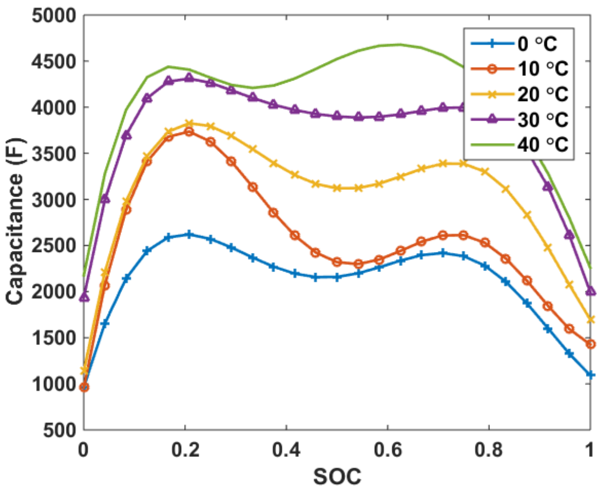
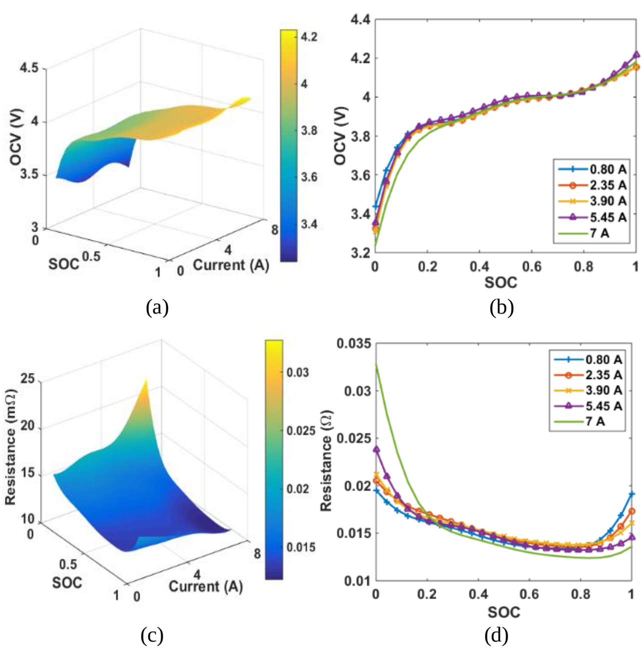


Fig. 9 Dependence of C_D on temperature

E. *Dependence of Parameters on Current Rate*

Current rate is another factor that affects the electrochemistry within lithium-ion batteries, and subsequently the electrical behavior. In the electrolyte, high rate discharge causes strong concentration polarization, resulting in highly nonlinear behaviors [8]. For the charge transfer kinetics, different current rates lead to different interfacial behaviors as mentioned before. As for OCV, the difference may be attibuted to different electrode reactions because of the concentration of the active materials. In summary, model parameters are obtained considering different current rates as shown in Fig. 10.

The profiles show that the model parameters don't change much with current rate when the SOC is between 30% and 70%. But when the SOC goes beyond 80% or below 20%, it will make a notable difference to parameters. The results in [28] show similar conclusions.



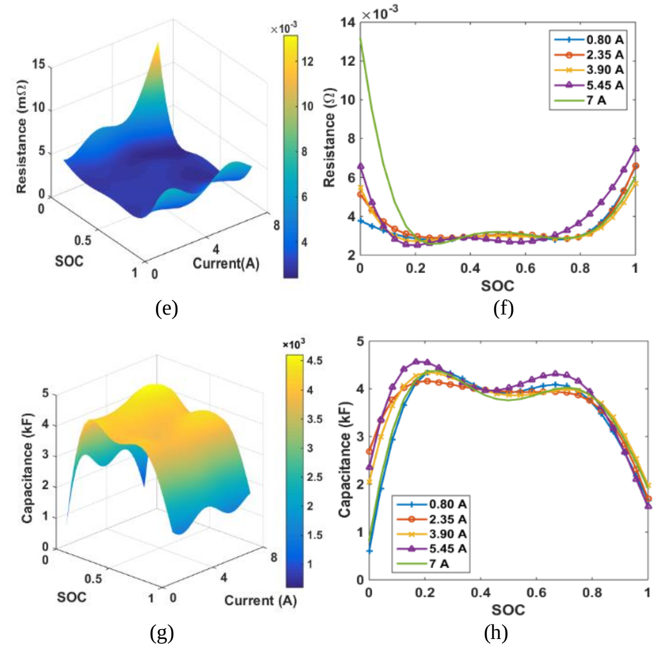


Fig. 10 Dependence of (a)(b) U_{OC} (c)(d) R_O (e)(f) R_D (g)(h) C_D on current rate

Conclusively, the dependence of these parameters on temperature and current rate and consequently changes in heat generation and transient response time is summarized in Figure 11. The simulation in the next section will follow this relationship.

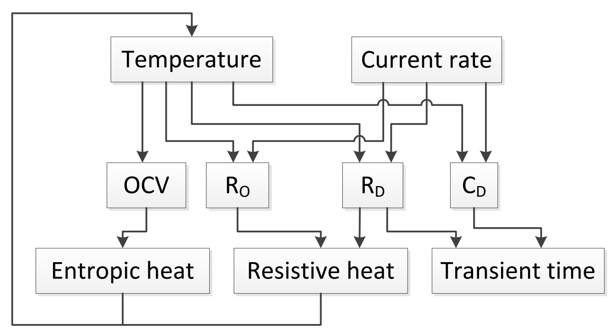


Fig. 11 Dependence of model parameters

IV. Experiment on Drive Cycles

A. *Dynamic Behavior of the Cell under Charge-Depleting Profile*

After the model parameters are estimated above, they will be used to predict the battery performance under different drive cycles. Since the charge-depleting (CD) mode is one of the operating modes for plug-in hybrid EVs defined by the United States Advanced Battery Consortium (USABC), it will be used for the validation of the electrical performance of the cell.

The USABC defines two operating modes for plug-in hybrid EVs, charge-depleting (CD) and charge sustaining (CS) [33]. In CD mode, the vehicle is powered by the electric motor and energy storage system. Since the vehicle is dependent on the battery pack, the SOC of the battery pack may fluctuate, while the overall trend is decreasing. When the SOC of the battery reaches a predefined minimum value, the vehicle would turn to

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CS mode, in which the SOC of the battery is kept relatively constant. Fig. 12 shows the current profile and its voltage response for the CD test.

The maximum discharging current in this case is about 7.4 A. The operating temperature is kept at 30 ºC. After 360s test, the discharged capacity is about 0.1 Ah and the SOC decreased by 6 %. During the test, the deviation between the modeling and experiment is within 1.5%. Overall, the model can predict the battery electrical behavior. The related calculation is given by:

$$\text{Deviation} = \frac{V_t - \widetilde{V_t}}{V_t} \tag{13}$$

where V_t is the experimental value and $\widetilde{V_t}$ denotes the value from the model. Notably, there are some spikes of errors during transients. One possible reason is that the effect of current direction is not included in the model. Since the porosity of electrodes varies during the charge and discharge, model parameters will change accordingly and affect the battery performance.

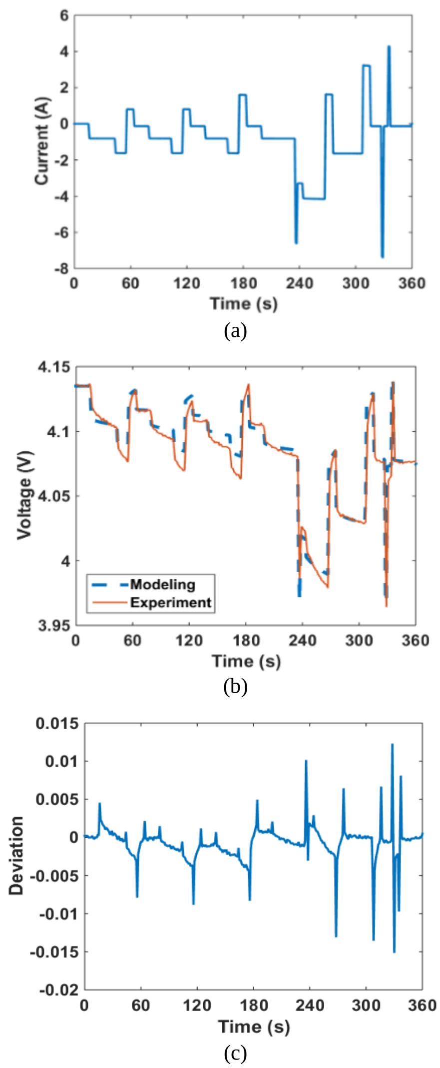


Fig. 12 (a) Current, (b) Voltage and (c) Deviation between the modeling and experiment of the CD test

B. Thermal Responses under Different Drive Cycles

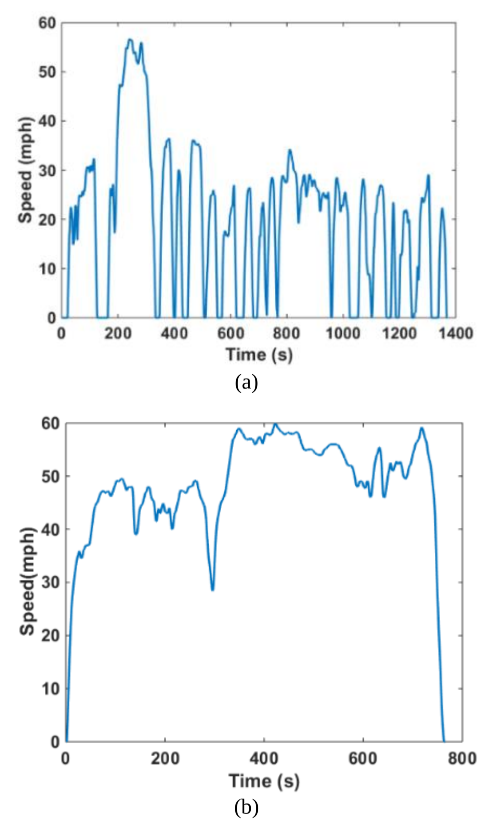
The thermal behavior of a LIB is not only affected by the parameters inside the battery, but also influenced by the external factors. When an electric vehicle is driving in different traffic situations, the LIBs inside will face different drive cycles. In this section, thermal behavior of LIBs will be studied using different drive cycles, including UDDS, HWFET and US06. These profiles are defined in the EPA Federal Test Procedure for the city driving conditions, by the US Environmental Protection Agency. Fig. 13 show the speed profiles of the drive cycles. Then the power profiles are calculated by the equations below.

$$P = \frac{V}{1000\eta_M}\left[Mg(\mu cos\alpha + sin\alpha) + \frac{1}{2}\rho A_f C_d V^2 + M\frac{dV}{dT}\right] \tag{14}$$

The parameters in the equations are listed in Table III. Considering the vehicle is running on level ground ($\alpha = 0$), the power profile (P) can be calculated accordingly. In summary, an overview of the power profiles of three drive cycles is presented in Table IV.

TABLE III
Specifications and operation conditions of the electric vehicle

Quantity	Symbol	Value
Mass	M	1243 kg
Frontal area	A_f	1.746 m^2
Air friction coefficient	C_d	0.26
Rolling resistance	μ	0.01
Air density	ρ	1.16 kg/m^3
Gravity	g	9.81 m/s^2
Wheel radius	R_{tire}	0.31 m
Efficiency	η_M	0.98



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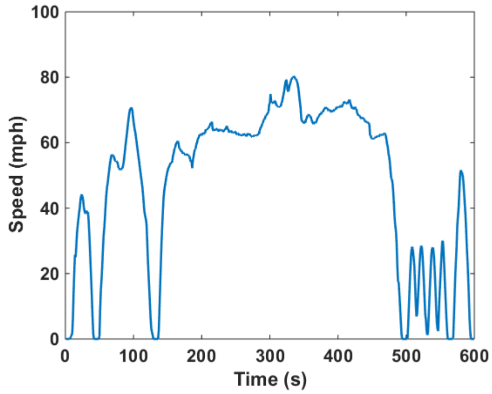


(c)

Fig. 13 Speed profiles of drive cycles (a) UDDS; (b) HWY; (c) US06

TABLE IV

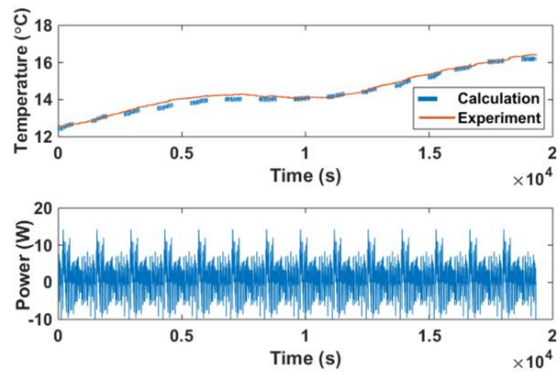
Overview of the power profiles of three drive cycles

Quantity	**UDDS**	**HWY**	**US06**
Time duration (s)	1370	765	600
Average power (W)	0.95	3.06	4.15
RMS power (W)	3.21	4.32	9.37
Peak power (W)	14.03	11.41	34.06
Standard deviation of power	6.14	3.05	8.41
Charge capacity of one cycle (Ah)	0.06	0.02	0.06
Discharge capacity of one cycle (Ah)	0.16	0.19	0.23

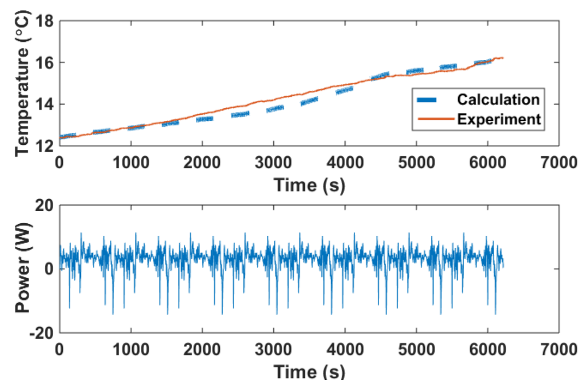
Since these drive cycles require different power and energy, they are scaled down using the same scaling factor, considering the fact that the number and configuration of batteries in an electric vehicle is fixed. For all the tests, the battery cell starts at the voltage of 4.11 V, runs the repeated test profiles until the terminal voltage hits 3V. The number of cycles for UDDS, HWY and US06 are 14.1, 8.1 and 8.5, respectively. In terms of thermal condition, all the tests start at around 12 ^{o}C and are kept in the adiabatic environment such that all the heat generation comes from the battery and heat dissipation is minimized. Finally, the power and temperature profiles of three drive cycles are presented in Fig. 14.

Overall, the final temperature under three drive cycles goes around 16 ^{o}C. However, considering the time duration of three cycles, the temperature rise under the US06 test is faster than that under the HWY test and the UDDS test is the slowest. One possible reason is that the RMS power of US06 test is the highest, then it is the HWY test and the UDDS test shows the lowest RMS power. Assuming that the variation of the terminal voltage is negligible, the power profile represents the current profile for the battery cell. Thus, the RMS power indicates the RMS current going through the battery cell, generating the corresponding amount of heat under different drive cycles. In addition, the peak power and the standard deviation of the power for the US06 profile are both the highest, meanwhile for one cycle, the US06 profile charges and discharges the most energy to and from the cell. In summary, the US06 profile shows the most aggressive power requirement, consequently leading to the highest temperature rise.

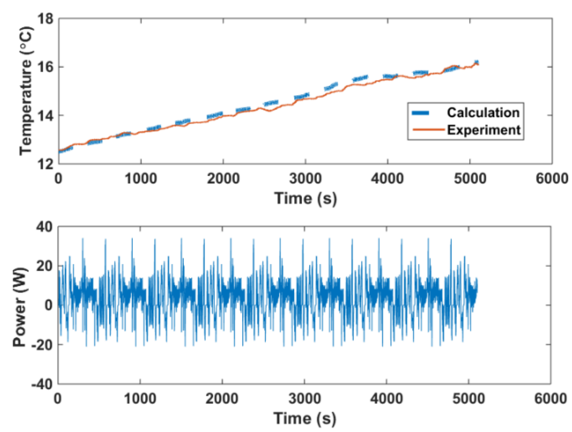
Another observation is that there is a period in the UDDS profile, which displays no temperature rise, even slight temperature decrease. One possible explanation is that the entropy effect of the cell overrides the resistive effect since the RMS power of the UDDS profile is small, which means the cell absorbs more heat than it generates. This is actually conducive to releaving the thermal stress in the battery. If the state of the cell can be kept within the level that is shown in Fig. 14 (a) as long as possible, the degradation effects and potential hazard caused by temperature can be possibly relieved.



(a) UDDS



(b) HWY



(c) US06

Fig. 14 Power and temperature profiles of (a) UDDS; (b) HWY; (c) US06

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V. Conclusion

This paper discusses in detail the dependence of the equivalent circuit model parameters on temperature and current rate. The experimental results show that, based on Arrhenius Law, battery temperature has a profound impact on model parameters. Moreover, current rates make a difference to the model parameters due to concentration dynamics. Although the experimental results have been supported by the previous report, more detailed explanation regarding the dependence of model parameters on current rate can be explored. Another limitation is that only one chemistry is used in this paper. The universal validity of the proposed modeling can be replicated and elaborated in the further work.

In addition, the dependence of the model parameters has been applied in the first order equivalent circuit model and used to validate the electrical and thermal behavior of the battery cell under different drive cycles. According to the analysis on the power profiles of the drive cycles, the corresponding thermal behavior has been explained. Apparently, a drive cycle with the aggressive power requirement generates heat fast and creates the potential thermal issues including hazard and degradation, whereas a mild drive cycle will relieve the thermal stress.

Appendix

The numerical relationship depicting the effects of the temperature and current rate is described by the polynomial as follows:

$$z = f(SOC, x) = \sum_{i,j} p_{ij} SOC^i x^j \quad (15)$$

where z represents the battery internal parameter, p_{ij} is the coefficient for each term and x depicts temperature or current rate for different cases.

In dependence of battery parameters on temperature, the current rate is fixed at 4A. Thus, the relationship in Fig. 6 showing the relationship between OCV, SOC and temperature can be expressed by (15) and the coefficient p_{ij} can be summarized in TABLE V.

TABLE V
Coefficients in the relationship between OCV, SOC and temperature

i \ j	0	1	2
0	3.442	−0.02671	0.002
1	5.229	0.04802	−0.005
2	−21.52	0.02662	0.002
3	42.52	−0.07104	0
4	−39.47	0.02891	0
5	13.96	0	0

Similarly, table VI to VIII show the coefficients in the relationship of Fig. 7-9.

TABLE VI
Coefficients in the relationship between R_O, SOC and temperature

i \ j	0	1
0	0.09076	−0.004391
1	−0.07886	0.002632
2	0.1334	−0.004534
3	-0.1412	0.004071
4	0.03238	−0.002017
5	0.03727	0

TABLE VII
Coefficients in the relationship between R_D, SOC and temperature

i \ j	0	1
0	0.0076	0
1	-0.03826	0.0006282
2	0.1562	-0.004384
3	-0.2136	0.005677
4	0.08269	-0.001569
5	0.01304	0

TABLE VIII
Coefficients in the relationship between C_D, SOC and temperature

i \ j	0	1	2
0	951.6	41.40	5.505
1	38540	-2321	94.42
2	-154200	3872	-106.3
3	277100	-2382	0
4	-226900	0	0
5	66200	0	0

In dependence of battery parameters on current rate, the temperature is fixed. The relationship in Fig. 10 can also be expressed by (15) and the coefficient p_{ij} can be summarized in TABLE IX to XII.

TABLE IX
Coefficients in the relationship between OCV, SOC and current rate

i \ j	0	1	2
0	3.506	-0.07214	-0.01469
1	4.771	0.6879	-0.1624
2	-22.97	-0.5837	0.1571
3	49.20	-0.2106	0
4	-47.19	0.2123	0
5	16.85	0	0

TABLE X
Coefficients in the relationship between R_O, SOC and current rate

i \ j	0	1
0	0.01648	0.003693
1	-0.01619	-0.01420
2	0.8353	0.01795
3	-0.2070	-0.02318
4	0.2141	0.01316
5	-0.07114	0

TABLE XI
Coefficients in the relationship between R_D, SOC and current rate

i \ j	0	1
0	0.004093	-0.0005652
1	-0.01817	0.0008498
2	0.08438	0.01249
3	-0.1831	-0.02313
4	0.1721	0.01326
5	-0.05401	0

TABLE XII
Coefficients in the relationship between C_D, SOC and current rate

i \ j	0	1	2
0	-1368	4220	-1730
1	40370	-12450	4030
2	-138400	7881	-2331
3	251800	3452	555.6
4	-220700	-3803	0
5	70130	0	0

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